KEYSTRA ENERGY GROUP

SOLUTION 3: SUPPLY CHAIN RISK AND RESILIENCE

Predictive Early Warning and Risk Management System

Portfolio Project | Proof of Concept

Data Source: Data Source: Synthetic (Python Engineered)

PROJECT OVERVIEW

The previous solution, Pipeline Integrity & Operational Risk Intelligence, established an asset-level view of operational risk by examining how failures within critical infrastructure can affect safety, reliability, and business continuity. The next step is to examine the dependencies that exist before risk reaches the asset.

Keystra's operational reliability depends on a supply network that connects suppliers, procurement, logistics, inventory, materials, and critical assets. Disruption at any point in this chain can affect maintenance activities, asset availability, production, and financial commitments. Supply Chain Risk & Operational Resilience was developed to identify and assess these dependencies from a risk perspective. The solution combines supplier performance, predicted failure probability, material criticality, asset dependency, business-unit exposure, downstream impact, and financial exposure to determine where supplier disruption could create the greatest business consequence. The focus is therefore not simply on identifying underperforming suppliers, but on understanding which supplier risks could materially affect the business, how far the impact could propagate, and where intervention should be prioritised.

BUSINESS CHALLENGE

For an integrated energy business, supplier risk can quickly become operational risk. A supplier failure may delay the delivery of critical materials, affect maintenance activities, restrict asset availability, and ultimately disrupt business operations. However, supplier performance indicators alone do not provide enough context to determine the significance of that risk. Two suppliers with similar performance may represent very different levels of business exposure depending on the materials they provide, the assets they support, the business units they reach, and the financial commitments attached to them.

The challenge was therefore to move beyond "which suppliers are risky?" to determine "which supplier risks could have the greatest consequence for Keystra?"

This required connecting supplier risk with the wider operational and financial dependency network, enabling the business to identify emerging risks, understand potential downstream exposure, and prioritise management action before supplier disruption becomes an operational event.

DATASET AND DATA PREPARATION

The analysis uses a synthetic supply chain dataset engineered specifically for Keystra Energy Group to simulate realistic supplier, procurement, inventory, quality, and operational dependencies. The data was designed to provide a connected view of the supply chain, linking suppliers → materials → procurement → inventory and quality → assets → business units. Existing Asset and Business Unit structures from Solution 1 were retained, while the supplier, material, procurement, quality, inventory, and risk-related datasets were engineered specifically for this solution.

The analytical period covers 1 January 2022 to 31 July 2026, providing the historical depth required to analyse supplier behaviour and develop the 90-day supplier failure prediction. The resulting data foundation was validated for data quality, relationship integrity, temporal consistency, and predictive leakage, creating a reliable basis for the subsequent risk investigations and predictive analysis.

PREDICTIVE SUPPLIER RISK AND OPERATIONAL EARLY WARNING ANALYSIS

Once the supplier, procurement, delivery, material and operational data had been prepared, the analysis moved from understanding what had already happened to asking a more forward-looking question:

Could a supplier's behaviour today provide an early indication that it may become materially unreliable in the future?

To answer this, Keystra created a daily supplier view covering all 90 suppliers from January 2022 to July 2026. The analysis considered recent fulfilment and delivery performance, outstanding orders, purchasing activity, and changes from each supplier's normal behaviour. A clear definition of supplier failure was then required.

The initial definition classified 93.85% of observations as failures because individual issues such as late deliveries or outstanding orders could trigger a failure label. This was too broad. The definition was therefore refined so that a supplier was considered to have experienced material failure only when performance deteriorated across at least two independent operational areas within the following 90 days. This reduced the failure rate to 7.79%, with 10,824 failures across 138,953 usable supplier observations.

Keystra then established a five-stage supplier risk journey:

Stable → Watch → Deteriorating → High Risk → Failure

Risk scores increased progressively across these stages, from 0.00 for Stable to 19.73 for Failure. All 90 suppliers reached at least the Deteriorating stage, 88 reached High Risk or above, and 80 eventually reached Failure. Importantly, failure was generally preceded by measurable deterioration, providing evidence that supplier distress could be identified before material failure occurred.

The predictive dataset brought together 54 indicators covering supplier performance, fulfilment, delivery behaviour, purchasing activity, outstanding orders, risk condition and changes from historical behaviour. Future failure outcomes were kept separate to prevent future information from influencing the prediction. A Random Forest model was first established as the baseline. On the unseen test period, it achieved 86.13% precision, 66.45% recall and 75.02% F1-score, detecting 4,613 of 6,942 future failures but missing 2,329. Threshold testing showed that performance was also affected by changing supplier conditions. The failure rate increased from 2.07% in training to 8.98% in validation and 32.89% in the final test period, demonstrating that the later supplier environment was substantially more failure-prone.

Keystra therefore compared alternative approaches. Histogram Gradient Boosting produced the strongest validation performance, with a PR-AUC of 0.9482 compared with 0.8795 for the original Random Forest. The final model, using a validated 0.30 warning threshold, achieved:

METRIC	BASELINE RANDOM FOREST	FINAL MODEL
Precision	86.13%	95.27%
Recall	66.45%	87.05%
F1-Score	75.02%	90.97%
PR-AUC	0.8878	0.9721
ROC-AUC	0.9598	0.9893

The final model detected 6,043 of 6,942 future failures, missing 899 and generating 300 false warnings. The strongest predictive signals were concentrated around recent supplier behaviour, particularly rolling purchase-order activity, fulfilment risk, recent fulfilment rates, partial fulfilment, outstanding quantities, delivery delays and deviations from historical performance.

This demonstrated an important finding: the system was not simply identifying suppliers that had performed badly in the past. It was identifying changes in supplier behaviour that could indicate emerging unreliability.

However, predicting that a supplier may fail was only the beginning. A probability such as 90% or 99% does not, by itself, tell Keystra how serious the risk is or what could be affected. This led to the next stage: turning supplier predictions into actionable risk warnings.

TURNING SUPPLIER PREDICTIONS INTO ACTIONABLE RISK WARNINGS

The early-warning layer combined predicted failure probability with the supplier's current risk condition, deterioration signals, exposure and dependency information. Using the latest modelling date of 30 July 2026, 88 suppliers had sufficient current information

for assessment. Applying the validated 0.30 threshold identified 63 suppliers requiring attention. The system then added explainability, evaluating 24 potential risk drivers including delivery delays, partial fulfilment, outstanding quantities, order activity, deterioration from historical behaviour and problems across multiple operational dimensions.

This allowed Keystra to distinguish between suppliers that were risky for different reasons. For example, S035 had a 97.68% predicted failure probability and was already in a Failure state, with deterioration across three operational dimensions. S025 had an even higher 99.31% probability, but its dominant signals were specifically related to worsening partial fulfilment. The system therefore moved beyond:

"This supplier is risky."

to:

"This supplier is risky because these aspects of its behaviour are deteriorating."

The next question was even more important: If this supplier becomes unreliable, how much does it actually matter to Keystra?

Connecting Supplier Risk to the Wider Supply Chain

Risk was therefore propagated through the dependency chain:

Supplier → Material → Asset → Business Unit

The analysis covered 90 suppliers, 60 materials, 13 assets and all five Keystra business units, with 407 unique supplier-material-asset relationships. The network represented 622,917 outstanding units and approximately $4.05 billion in outstanding procurement exposure. The propagated exposure reconciled exactly with the underlying procurement exposure, producing a reconciliation ratio of 1.00 with no orphan dependency links. This allowed Keystra to distinguish between supplier risk and business exposure to supplier risk.

For example, SunCore Energy Technologies had approximately a 99% predicted failure probability and a propagation score of 94.92, with dependencies across three materials, five critical assets and four business units. DeltaFlow Technologies carried approximately $456.2 million in exposed outstanding value. These cases demonstrated that supplier risk does not stop at the supplier. The consequences can move through materials, assets and business units.

From Risk Assessment to Intervention

The intervention engine combined predicted failure probability, supplier priority, financial exposure, critical material dependency, asset dependency and potential business impact. This resulted in:

* 16 Immediate intervention

* 36 High-priority intervention

* 27 Moderate intervention

* 11 Monitor

The framework therefore recognised that probability alone should not determine priority. A supplier with a lower probability of failure can still require significant attention if its potential consequences are substantial.

The Final Early-Warning Layer

The final early-warning engine translated the combined risk assessment into severity, risk type, recommended response and management escalation. Across the 90 suppliers, the final warning distribution was:

SEVERITY	SUPPLIERS	SHARE
Critical	10	11.11%
High	26	28.89%
Moderate	35	38.89%
Low	17	18.89%
No Current Warning	2	2.22%

Critical warnings required action within 0–7 days, while High warnings required action within 14 days.

The system also differentiated between Predictive Supplier Failure Alerts, Emerging Supplier Risk Alerts, Supply Chain Exposure Alerts, combined Failure and Exposure Alerts, and general Supplier Risk Monitoring. The two suppliers without a current predictive warning were deliberately not treated as risk-free. Their status meant that there was insufficient current predictive evidence to issue a stronger warning.

The final analytical chain therefore became:

Supplier Risk → Predicted Failure → Risk Explanation → Dependency Exposure → Operational Impact → Intervention Priority → Early-Warning Severity → Recommended Response

THE FINAL SUPPLIER RISK REGISTER: FROM 90 SUPPLIERS TO CLEAR MANAGEMENT PRIORITIES

Everything developed throughout the Keystra supply-chain risk journey ultimately comes together in the Final Supplier Risk Register. The purpose of the register is not simply to rank 90 suppliers from highest to lowest risk, it is the final expression of the wider risk framework built throughout the analysis. Each supplier has been assessed based on the likelihood of future deterioration, the organisation’s exposure to that supplier, the materials and assets connected to it, the potential for risk to propagate through the supply chain, and the level of management attention required. The result is a complete and explainable classification of the supplier population. Every supplier now has a defined risk position, Early Warning Score, intervention priority and primary risk driver.

RISK CLASSIFICATION	SUPPLIERS	SHARE
Critical Early Warning	1	1.11

Risk Classification	Suppliers	Share
Critical Early Warning	1	1.11%
High Early Warning	10	11.11%
Emerging Risk	21	23.33%
Watchlist	34	37.78%
Low Current Risk	24	26.67%

The distribution reveals significant risk concentration. Only 11 suppliers (12.22%) fall within the Critical or High categories, yet they represent approximately $1.70 billion, or 42.01%, of the $4.05 billion total outstanding exposure. This means Keystra does not need to treat all 90 suppliers equally; a relatively small group carries a disproportionate share of potential business exposure.

S004 — Sterling Industrial Equipment Ltd. ranks highest, with an Early Warning Score of 77.5/100, a 92% predicted failure probability, 73.14/100 propagation score, and approximately $1.258 billion in outstanding exposure. Its primary risk driver is Financial Exposure, resulting in an Immediate intervention priority. Its position demonstrates why supplier risk cannot be assessed using failure probability alone: the final ranking considers both likelihood and consequence.

The 10 High Early Warning suppliers also illustrate this distinction. Some are primarily driven by Business Unit Reach, others by Supplier Failure Probability, while ProSense Industrial Controls Ltd. is driven by Critical Material Exposure.

Several suppliers have predicted failure probabilities of 97–99%, including Process Materials Solutions, Continental Industrial Services, Nexus Industrial Supply, SunCore Energy Technologies, Advanced Process Instruments, and Crown Electrical Technologies. However, their final risk positions differ because the potential consequences of their disruption are not the same.

SunCore, for example, combines a 99% failure probability with a 94.92/100 propagation score, while ProSense has a lower 64% probability but a 81.63/100 propagation score and approximately $58.77 million exposure.

The 21 Emerging Risk suppliers represent an important early-intervention layer. Examples include VoltEdge, driven by Critical Asset Exposure, and DeltaFlow, driven by Downstream Propagation with approximately $456.2 million in outstanding exposure. These suppliers may not yet require the highest level of intervention, but their developing risk signals warrant investigation and mitigation before conditions escalate.

The remaining 34 Watchlist suppliers require enhanced monitoring, while the 24 Low Current Risk suppliers remain under routine monitoring. Importantly, a Watchlist classification does not mean no risk, and a low-risk score does not guarantee future stability. It simply reflects the strength of current risk signals and the level of attention warranted at that point in time. Overall, the supplier population carries approximately $4.05 billion in outstanding exposure, with an exposure-weighted risk score of 57.97/100. The dominant risk driver is Downstream Propagation, reinforcing a central finding of the analysis:

Supplier risk does not stop at the supplier.

A supplier may support a material, the material may support an asset, and the asset may support a business unit. The final register therefore connects supplier risk → material dependency → asset exposure → business impact → intervention priority.

The outcome is more than a supplier ranking. It provides Keystra with a structured risk hierarchy showing who is becoming unreliable, why they are at risk, what could be affected, how significant the exposure is, and how urgently the organisation should respond.

PROJECT LIMITATIONS

Although the Supply Chain Risk & Operational Resilience Early-Warning System demonstrates a complete risk analytics framework, several limitations should be recognised:

1. **Synthetic data: The dataset was engineered specifically for Keystra Energy Group to simulate realistic supply-chain relationships. While designed to reflect plausible operational patterns, it does not represent actual Keystra supplier or procurement records. The findings should therefore be interpreted as a proof of concept rather than direct evidence of real supplier risk.**
2. **Limited external variables: The analysis primarily uses internal supply-chain signals such as purchase orders, deliveries, fulfilment, quality events, inventory, supplier performance and dependency relationships. External factors such as geopolitical**

events, inflation, exchange-rate movements, market disruptions, weather, regulatory changes and supplier financial health were not incorporated.

3. **Predictive horizon: Supplier failure was defined around a 90-day forward-looking window. This provides a practical early-warning horizon but may not capture longer-term structural supplier risks or disruptions that develop beyond 90 days.**
4. **Historical and engineered patterns: Because the data is synthetic, the predictive model identifies patterns within the behaviours represented in the engineered dataset. Performance on real-world data may differ, particularly when faced with new failure patterns, changing supplier behaviour or unexpected disruptions.**
5. **Temporal drift: The analysis identified changes in failure rates across the training, validation and test periods. This indicates that supplier risk conditions can change over time and that the predictive model would require ongoing monitoring, recalibration and periodic retraining in a real production environment.**
6. **Model limitations: Even with strong predictive performance, the model does not eliminate false warnings or missed failures. Predictions should therefore support, rather than replace, supplier investigation and professional risk assessment.**
7. **Data quality and availability: A production implementation would depend on consistent, timely and accurate procurement, supplier, inventory, quality and operational data. Missing, delayed or inconsistent records could reduce the reliability of risk scores and early warnings.**

These limitations do not invalidate the framework. Instead, they define the conditions required to move the solution from a portfolio proof of concept toward a production-ready supply-chain risk monitoring and early-warning system.

PROJECT ARTEFACTS & FULL ACCESS

Available Artefacts:

Full Project Report | Executive Summary | Python Code | Engineered Datasets

These are all available through the link below

Github:

https://github.com/winnerEM/Keystra-Energy-Solution-3_Supply-Chain-Risk-and-Resilience

Also accessible on google drive:

https://drive.google.com/drive/folders/1FAM1VRI_G8APz_9uBdB9tisQdKKmjtch?usp=sharing